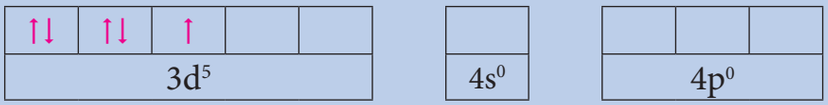
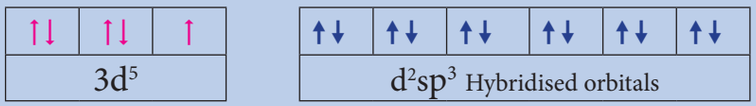
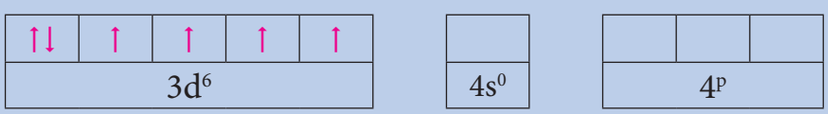
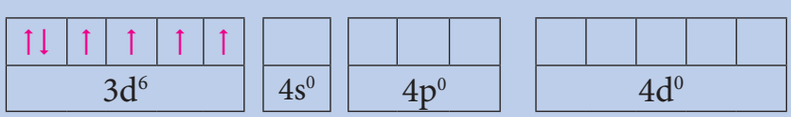
Complex	$[Fe(CN)_6]^{3-}$
Outer orbitals of metal atom/ion in presence of ligands	
Hybridisation	Coordination number - 6 Hybridisation - d^2sp^3
Hybridised orbitals of the metal ion in the complex	
Geometry	Octahedral In this complex inner d orbitals are involved in hybridisaion and hence the complex is called inner orbital complex
Magnetic property	No. of unpaired electrons = 1; Hence paramagnetic
Magnetic moment (Using spin only formula)	$\mu_s = \sqrt{n(n+2)} = \sqrt{1(1+2)} = 1.732$ BM

Illustration 4

Complex	$[CoF_6]^{3-}$
Central metal atom and its outer electronic configuration	Co^{3+}: $3d^6$, $4s^0$
Outer orbitals of metal atom/ion	
Nature of ligand	F^- Weak field ligand and hence no pairing of 3d electrons in the metal
Outer orbitals of metal atom/ion in presence of ligand	
Hybridisation	Coordination number - 6 Hybridisation - sp^3d^2

XII U5 Coordination jagan.indd 151 22-12-2022 19:51:04